

Black Hole Attack Detection in MANETs

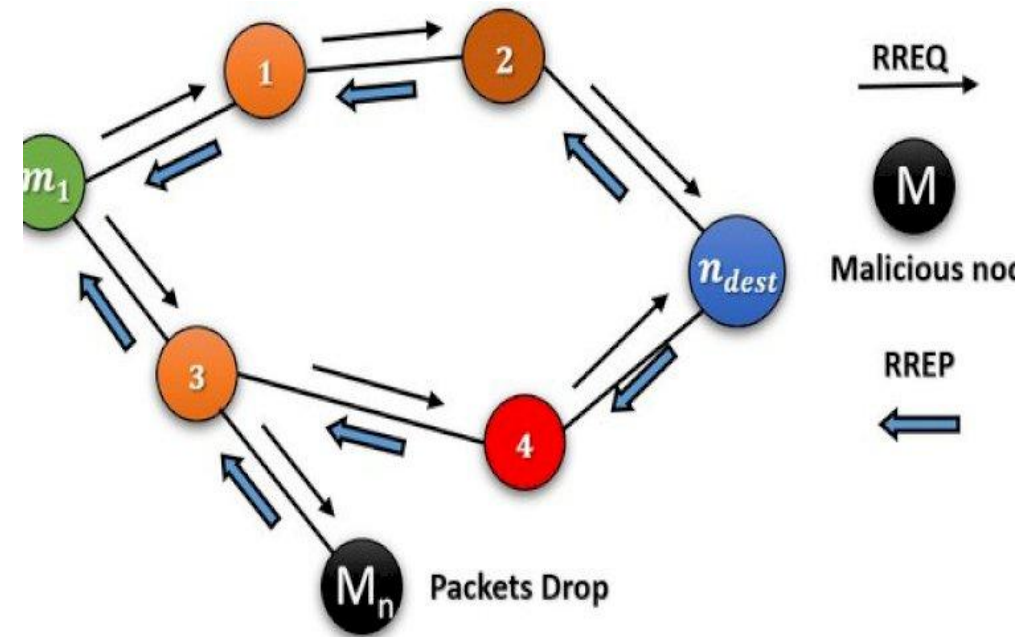
Team Members:

1. Mageshwaran M - 127158029
2. Dhiyaneshwar P V - 127158010
3. Yuvanesh Sasikumar - 127158061

Problem Statement

Mobile Ad-hoc Networks (MANETs) are decentralized and infrastructure-less, making them highly vulnerable to routing attacks.

- **Black Hole Attack:** A malicious node falsely advertises the shortest path to a destination during the route discovery process.
- **Impact:** Once the route is established, the node silently drops all data packets, causing 100% data loss and network disruption.
- **Challenge:** Detecting these nodes in real-time without compromising network performance is difficult due to the dynamic topology.



Literature Survey & Research Gap

Primary Reference

Abdallah et al. (2024)

International Information & Engineering
Technology Association (IETA)

Focus: Proposed the ADS-SVM model
(Anomaly Detection System using Support
Vector Machines).

Technique: Utilized a Radial Basis Function
(RBF) Kernel to handle non-linear data patterns.

Key Result: Achieved **99.96% detection
accuracy** in controlled simulations.

Identified Gap Analysis

While theoretical models achieve high
accuracy, current research lacks practical
implementation tools.

Modern Integration: Lack of Python-based,
lightweight implementations suitable for
modern dashboards.

Visualization: Absence of real-time visual
feedback for immediate threat assessment.

Proposed Methodology

1. Detection Engine (Backend)

We will implement the **Support Vector Machine (SVM)** algorithm with an RBF kernel, as recommended by Abdallah et al.
Features: Packet Delivery Ratio (PDR) and Throughput.

Training: Using the UNR-IDD dataset

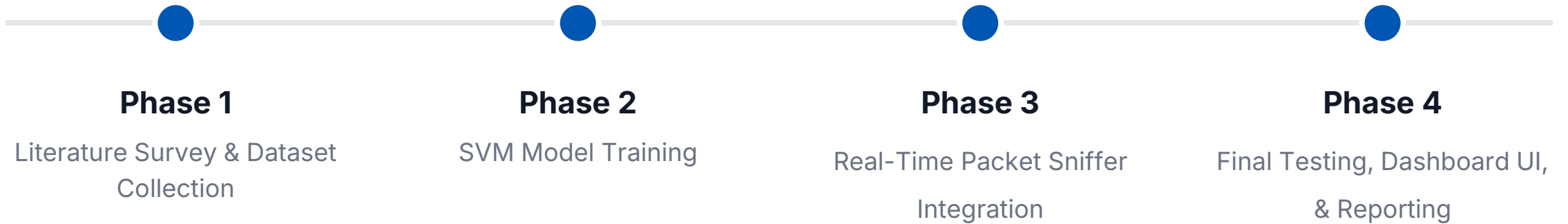
2. Real-Time Monitor (Frontend)

A live detection system bridging the gap between theory and practice.

Capture: Python **Scapy** library to sniff live network packets.

Visualization: **Streamlit** dashboard to display network health and flag malicious IPs instantly.

Project Timeline & Work Plan



Technical Implementation Progress

- **Dataset:** Transitioned from NSL-KDD (Generic) to **UNR-IDD (Network Intrusion Dataset)** for better SDN/MANET behavior relevance.
- **Core Model:** SVM with Radial Basis Function (RBF) Kernel.
- **Feature Engineering:** Focused on 13 behavioral features (Deltas, Throughput, and Flow entries) to eliminate data leakage found in cumulative packet counts

Experimental Results (Current Stage): Model accuracy of 75.37% achieved after removing leaky cumulative features and room to improve.

Accuracy: 75.37%					
	precision	recall	f1-score	support	
0	0.88	0.74	0.81	2526	
1	0.58	0.78	0.66	1132	
accuracy			0.75	3658	
macro avg	0.73	0.76	0.73	3658	
weighted avg	0.79	0.75	0.76	3658	

System Architecture and Progress (Prototype)

- **Sniffer Engine (Scapy):** Performs live capture of inbound (RX) and outbound (TX) traffic to detect packet absorption.
- **Real-time Logic:** Uses **Bi-directional Symmetry Analysis** to distinguish between high-bandwidth legitimate traffic (Speed Tests) and Black Hole attacks.
- **Dashboard (Streamlit):** A live interface displaying node identity (IP/MAC), estimated PDR, and real time analytics